Data set from a speed-incentivized associative reward learning task

This data set contains behavioural data from 79 right-handed healthy individuals, aged 18-40 years. The data set is subdivided into a pilot data set (N=20) and a main data set (N=59). In the study, participants performed a speed-incentivized associative reward learning task (see below). After the task, participants filled in a debriefing questionnaire. The data was acquired between March and December 2021 at the Translational Neuromodeling Unit (TNU) in Zurich, Switzerland.

Short description of the task

In the task, participants were required to learn the probabilistic association between two fractals and monetary rewards. Importantly, the underlying probabilistic associations in this experiment changed over time (volatility). Participants were asked to choose the fractal which they believed to be rewarded on a trial-wise basis which effectively corresponded to a (binary) prediction. Measured response times associated with these predictions are also contained in the data set.

The structure of a trial in the task looked as follows: Participants were presented a fixation cross in the middle of a computer screen. Below the fixation cross two fractals were shown that could be chosen, and a time bar was presented as well. A trial – during which a prediction was made (duration 1700 ms) – was always followed by an inter-trial interval (ITI, timing was jittered). At the beginning of the next trial, the outcome of the previous trial was revealed (correct prediction: coin, incorrect prediction: crossed coin, no/missing prediction: “Keine Taste gedrückt!”). Simultaneously with the onset of a new trial, the time bar started receding, signalling the available response time. The participant’s choice (predicted fractal) was indicated by a grey frame around the fractal and the filling level of the time bar indicated the relative amount of time required for prediction.

A customized payoff structure served to incentivize fast responses while still emphasizing the importance of correct predictions. Participants were informed about the payoff structure in advance. Specifically, correct trials were rewarded, with the amount of the monetary reward depending on the speed of the response. Incorrect trials never yielded any reward (independent of response speed). Block randomization across participants was used for the reward trajectories of the two fractals as well as the fractal positions (left/right).

For a more detailed description of the task including visualisations, the inclusion criteria and the data pre-processing, please refer to the corresponding analysis plan (https://doi.org/10.5281/zenodo.10669944).

Description of data included in this data set

The file *SPIRL_main_59sub_preprocessed_IncludedInAnalysis.mat* contains the main data set (N=59) in the form of a MATLAB struct with the following four fields:

- SPIRL (1x59 struct with 4 fields)
    - u (160x1 double) containing the binary input sequence of the task in contingency space. A 1 means that fractal A was rewarded in a given trial and fractal B yielded no reward, and vice-versa for a 0.
    - y (160x2 double) containing the responses from a participant. The first column contains the binary predictions (1=fractal A, 0=fractal B) while the second column contains measured log transformed response times (in ms).

    - yellow_left: binary variable indicating whether the yellow fractal was presented on the left side of the screen (1) versus on the right side (0). The green fractal was presented on the other side of the screen, respectively.
    - yellow_high: binary variable indicating whether the yellow fractal (1) was rewarded on the first trial or whether the green fractal (0) yielded a reward on the first trial.
- logRT_mat (160x59 double): containing a matrix of the log transformed response times (in ms) (row: trial, column: participant).
- traj (1x1 struct with 3 fields)
    - probs_green (160x1 double): containing the underlying probability structure of the experiment, which was used to generate the binary input sequence u.
    - u (160x1 double): containing the binary input sequence of the task in contingency space. A 1 means that fractal A was rewarded in a given trial and fractal B yielded no reward, and vice-versa for a 0.
    - block (1x1 struct with 2 fields)
        - code (160x1 double): containing integers from 1 to 13 indicating which phase of the experiment each trial belonged to.
        - length (1x13 double): containing integers indicating the number of trials in each phase of the experiment.
- trials (1x160 double): containing the trial numbers in ascending order.

The file *SPIRL_pilot_20sub_preprocessed_IncludedInAnalysis.mat* contains the pilot data set (N=20) in the form of a MATLAB struct with the same four fields as described above.

The two files *SPIRL_main_59sub_preprocessed_IncludedInAnalysis.json* and *SPIRL_pilot_20sub_preprocessed_IncludedInAnalysis.json* contain the same data as in the above described struct 'SPIRL' but in a standard text-based format (JSON). Note that all relevant information for the analysis is in the json files. However, the additional variables in the .mat files are used for visualization purposes.

Ethics approval

All participants gave written informed consent prior to data acquisition and were financially reimbursed for their participation. The study was approved by the Ethics Commission of ETH Zurich (ETH-EK-Nr. 2021-N-05).

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